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#include <iostream>
using namespace std;
double f(double x) {
    return x * x;
}
double simpsons38(double a, double b, int n) {
    if (n % 3 != 0) {
        cout << "n must be a multiple of 3. Adjusting n to the nearest multiple of 3." << endl;
        n += (3 - n % 3);
    }
    double h = (b - a) / n;
    double sum = f(a) + f(b);
    for (int i = 1; i < n; i++) {
        if (i % 3 == 0) {
            sum += 2 * f(a + i * h);
        } else {
            sum += 3 * f(a + i * h);
        }
    }
    return sum * (3 * h / 8);
}
int main() {
    double a, b;
    int n;
    cout << "Enter the lower limit (a): ";
    cin >> a;
    cout << "Enter the upper limit (b): ";
    cin >> b;
    cout << "Enter the number of subintervals (n): ";
    cin >> n;
    double result = simpsons38(a, b, n);
    cout << "The approximate integral from " << a << " to " << b << " is: " << result << endl;
    return 0;
}

```